DEEPAK KATUKURI

AI ENGINEER · AGENTIC AI · WORKFLOWS



CONTACT

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TECHNICAL SKILLS

Languages: Python, SQL, PL/SQL, Core Java, C++

AI Automation & Agents: LangGraph, multi-agent pipelines, prompt engineering, Claude/LLM orchestration, Chroma vector DB, semantic retrieval (sentence-transformers), Typer CLI

LLM & RAG: Retrieval-Augmented Generation, LangChain, hybrid retrieval (BM25 + dense, RRF), cross-encoder reranking, pgvector, embeddings, Ollama

Full-Stack: FastAPI, Flask, React, Vite, Tailwind CSS, PostgreSQL, MySQL, SQLAlchemy, Alembic, JWT, Recharts

ML & Data: Scikit-Learn, classification, ensembles, NumPy, Pandas, Matplotlib, Seaborn

Tools: Git/GitHub, pytest, Jupyter, Linux (Ubuntu)

EDUCATION

MSc Data Science
University of Europe, Germany
Sep 2025 – Present

BTech in Computer Science & Engineering
MLRIT, India
Jun 2018 – Jul 2022

CERTIFICATIONS

PL/SQL (Database Management)
Core Java (Software Development)
OneShield (P&C Insurance)

LANGUAGES

English – Fluent
German – A1/A2 (learning)

PROFILE

AI Engineer and Data Science MSc candidate, with 2.5 years of production software engineering behind the AI work. Built JobPilot, a 12-node multi-agent LangGraph pipeline that automates job-requirement analysis and CV drafting end to end, gated by a hard fact-validation check that blocks any unverified claim. Hands-on daily with prompting and LLM orchestration – hand-rolled a hybrid RAG retriever and validated it against a LangChain rebuild on a 55-question golden set. Structured, detail-oriented, and ships production software: runs a live full-stack health app today. Looking for an English-speaking working student (Werkstudent) role in Germany.

PROJECTS

JobPilot: Multi-Agent Job-Application Pipeline

Python · LangGraph · LangChain · Chroma · Typer · github.com/Deepak589/Job_hunt_Agentic_AI

- Automated the JD-to-tailored-CV workflow end to end by architecting a 12-node LangGraph state machine – requirement extraction, diagnosis, rewrite, fact validation, review and ATS scoring – orchestrating Claude through LangChain.
- Built a hard fact-validation gate that blocks any CV claim lacking evidence in a structured profile store, backed by a Chroma-based semantic evidence retriever over sentence-transformer embeddings: zero fabrication by design, not by review.
- Replaced subjective resume scoring with a deterministic ATS engine computing pass / fix / skip verdicts from counted, PDF-verified fact tables, covered by 84 pytest tests across 14 real and synthetic job-description fixtures.

RAG Pipeline: Hand-Rolled Retrieval, Validated Against LangChain at Scale

Python · LangChain · BM25 + dense hybrid · pgvector · Ollama · github.com/Deepak589/RAG_pipeline

- Built a hand-rolled hybrid retriever from scratch over 4 iterative stages: BM25 plus a dense bi-encoder, RRF fusion, cross-encoder reranking. Reached 0.833 recall@5 and 0.672 MRR on a 55-question labeled golden set before adopting any framework.
- Rebuilt the same retriever on LangChain using the identical corpus and labels, with the framework as the only changed variable. Caught a silent 0.06-recall regression from LangChain's default BM25 tokenizer and fixed it to 0.854 recall@5, so both implementations matched before scaling up.
- Scaled the corpus ~100x, from 7 to ~740 documents and ~3.5GB across arXiv, SEC filings, Wikipedia and scanned PDFs, using a resumable ingestion pipeline that routes by file type into a pgvector-backed dense store.

RA Nutrition & Lifestyle App: Shipped Full-Stack Product

React · FastAPI · PostgreSQL · Docker · live at ra-project-indol.vercel.app

- Deployed a live full-stack health app as two tiers: a React/Vite/Tailwind frontend on Vercel, and a FastAPI + PostgreSQL backend (4 logical schemas) running as a Docker container on Render, with JWT auth and token refresh.
- Built a rule-based recommendation engine on top of USDA FoodData Central, plus a weekly Recharts dashboard that turns raw meal and symptom logs into specific suggestions for the user.

EXPERIENCE

Software Engineer

Mar 2022 – Sep 2024

ValueMomentum · clients: Hiscox, Philadelphia Insurance

- Migrated full policy, claims, premium and customer history for Philadelphia Insurance's Surety Bonds line from a legacy policy admin system into OneShield PAS. Wrote and validated the PL/SQL migration scripts across customer, policy (PID), claims and premium tables.
- Root-caused a recurring data-integrity defect where claims data for pre-2019 Surety bonds under one bond type loaded empty. The OneShield object copy from the stored object to the loading object failed silently when another bond shared that bond type under a different bond ID. Kept records current with a weekly backfill script while a permanent fix was pending.
- Resolved 100+ production defects across two carrier platforms, Hiscox (P&C insurance) and Philadelphia Insurance (surety bonds). Turned underwriting and claims requirements into technical specs, and ran QA/UAT test cases and CI/CD deployments across the full policy lifecycle.

ML Intern

Jun – Jul 2019

SmartBridge × IBM

- Built a lung-cancer risk-prediction ensemble (Random Forest, KNN, Gradient Boosting, Neural Network) and deployed it on IBM Watson ML. Presented the results to non-technical reviewers with Matplotlib and Seaborn charts.